

Adding nrf21540 RF FEM (GPIO Mode) support into any existing project

Documented on: **21/05/26** by *Rohan Singh*

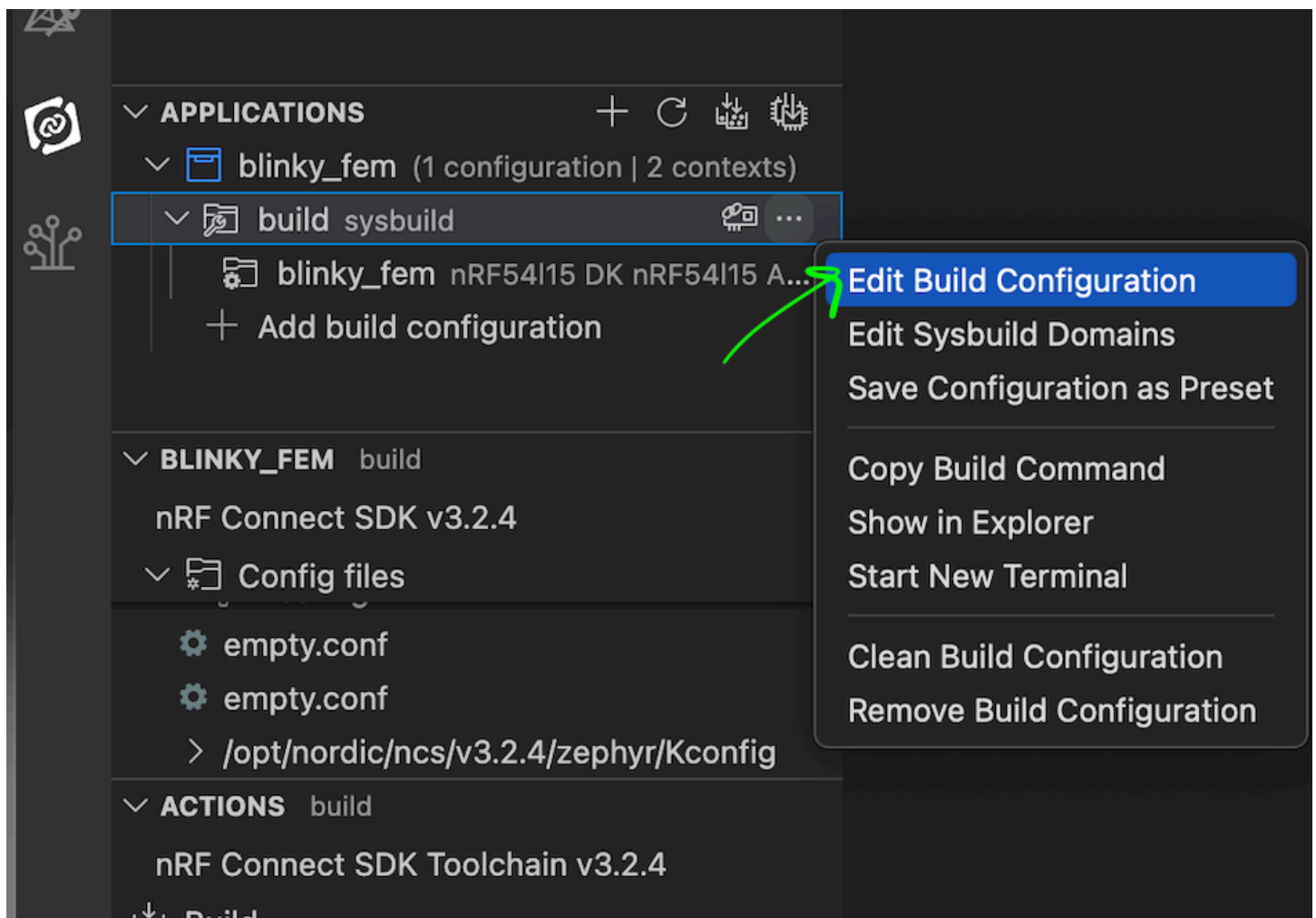
Final code (github):

[GitHub - angeryrohan/ble_with_fem: Taking 1M advertiser as the base, adding RF FEM Su...](#)

CONTEXT: There are 3 modes (GPIO, SPI, GPIO+SPI). This document guides on implementing the GPIO mode.

Steps:

1. Edit build configuration



2. Add CMake Argument: **-DSHIELD=nrf21540ek**

~/mlwb/blinky_fem/src/main.c

Edit Build Configuration (build)

SDK ?

nRF Connect SDK v3.2.4

Toolchain ?

nRF Connect SDK Toolchain v3.2.4

Board target ? ⚠

Revision ?

nrf54l15dk/nrf54l15/cpuapp

Revision

☒ Nordic SoC ☐ Nordic Kits ☐ Custom ☐ All

Base configuration files (Kconfig fragments) ?

Select...

Browse

Extra Kconfig fragments ?

Select...

Browse

Base Devicetree overlays ?

Select...

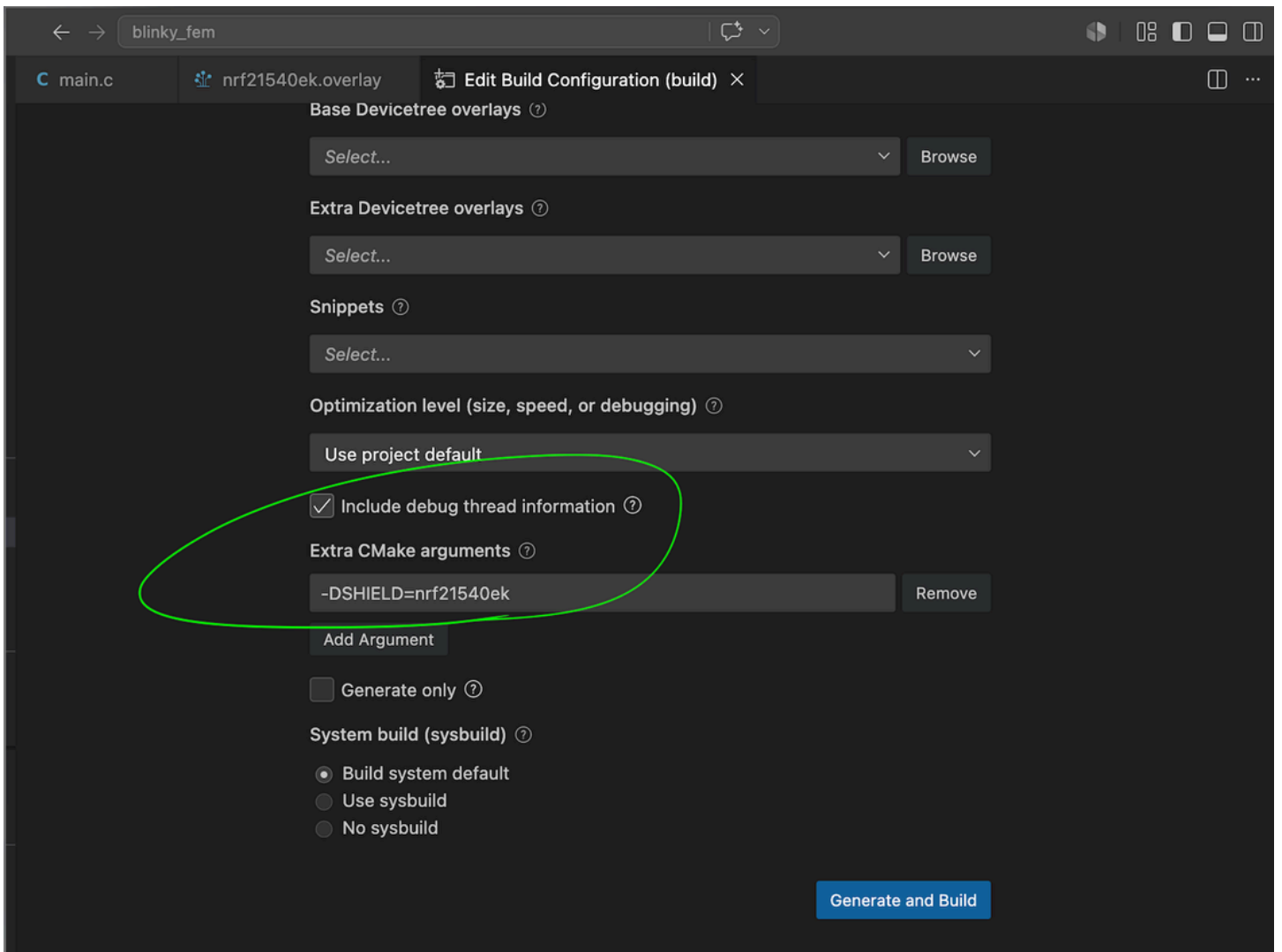
Browse

Extra Devicetree overlays ?

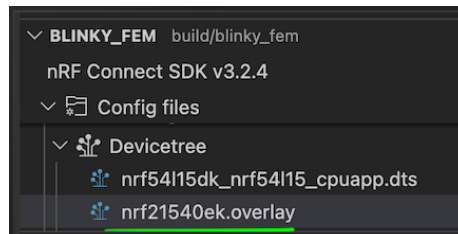
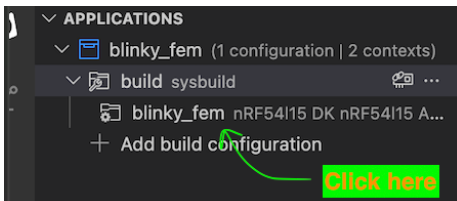
Select...

Browse

Scroll for more



3. Click in your build file → Open the nrf21540ek.overlay by clicking here!



4. Copy paste this code in nrf21540ek.overlay

CASE 1: you have connected the ANT_SEL pin of the NRF21530 RF FEM to a GPIO of microcontroller. Example:

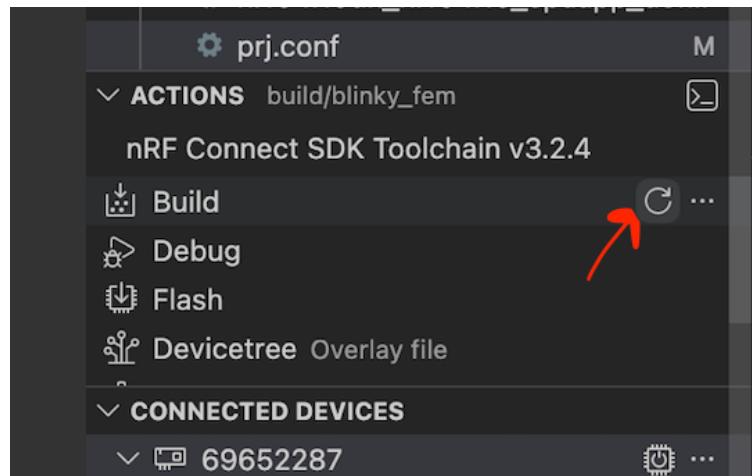

```
};
```

```
&radio {  
    fem = <&nrf_radio_fem>;  
};
```

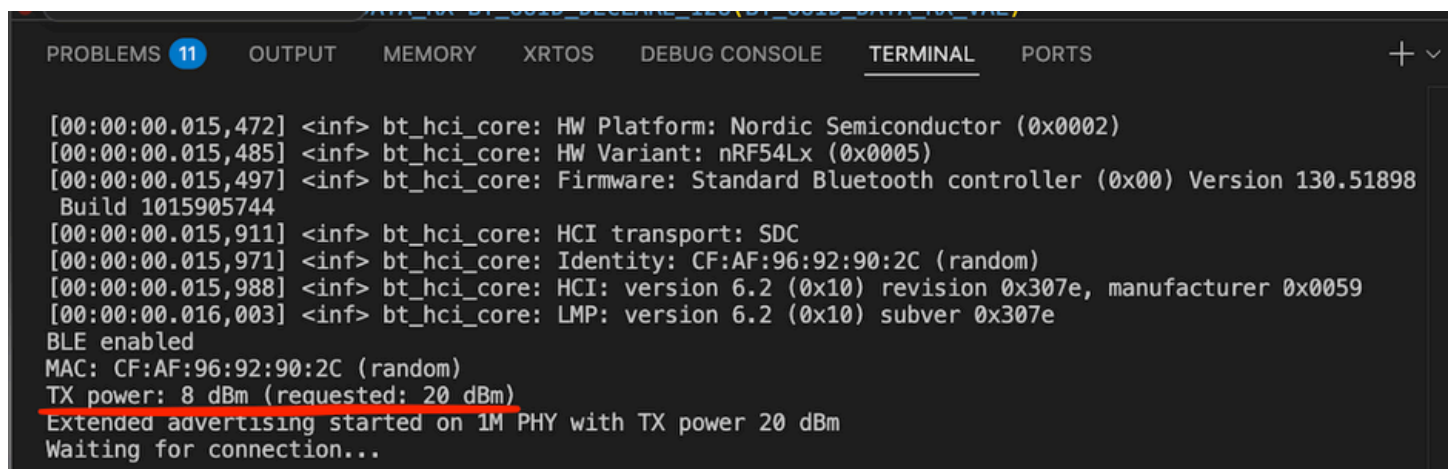
4. Copy paste this code in prj.conf (**NOTE: CONFIG_BT_CTLR_TX_PWR_ANTENNA** decides the TX Power, Input in dBm, max: 21dBm)

```
CONFIG_MPSL=y  
CONFIG_MPSL_FEM=y  
CONFIG_MPSL_FEM_NRF21540_GPIO=y  
CONFIG_MPSL_FEM_NRF21540_TX_GAIN_DB=20  
CONFIG_BT_CTLR_TX_PWR_ANTENNA=20
```

5. Perform a pristine build



BEFORE:



AFTER:

```
PROBLEMS 19 OUTPUT MEMORY XRTOS DEBUG CONSOLE TERMINAL PORTS + \n\n[00:00:00.015,475] <inf> bt_hci_core: HW Platform: Nordic Semiconductor (0x0002)\n[00:00:00.015,488] <inf> bt_hci_core: HW Variant: nRF54Lx (0x0005)\n[00:00:00.015,499] <inf> bt_hci_core: Firmware: Standard Bluetooth controller (0x00) Version 130.51898\nBuild 1015905744\n[00:00:00.015,885] <inf> bt_hci_core: HCI transport: SDC\n[00:00:00.015,946] <inf> bt_hci_core: Identity: CF:AF:96:92:90:2C (random)\n[00:00:00.015,964] <inf> bt_hci_core: HCI: version 6.2 (0x10) revision 0x307e, manufacturer 0x0059\n[00:00:00.015,981] <inf> bt_hci_core: LMP: version 6.2 (0x10) subver 0x307e\nBLE enabled\nMAC: CF:AF:96:92:90:2C (random)\nTX power: 20 dBm (requested: 20 dBm)\nExtended advertising started on 1M PHY with TX power 20 dBm\nWaiting for connection...
```

FAQ

GPIO Pin Selection

- The pins in the overlay (gpio1 6, gpio1 9, etc.) must match your physical wiring between the nRF54L15DK and the nRF21540 EK. These are not universal → verify against their schematic before copy-pasting
- ant-sel-gpios is optional – only needed if you're using antenna diversity (THE RF FEM automatically switches between antennas)

CONFIG values – when to change

- CONFIG_MPSL_FEM_NRF21540_TX_GAIN_DB=20 – 20dB is the nRF21540's fixed TX gain, don't change unless using a different FEM variant
- CONFIG_BT_CTLR_TX_PWR_ANTENNA=21 – **this is the main power controller**

FAQ to add

- "SPI mode vs GPIO mode – when do I care?" – GPIO mode is simpler but gives less runtime control; SPI mode lets you dynamically control gain. GPIO mode is fine for most BLE use cases